

DWMA: Autonomous Sensorimotor Structure Acquisition, Multi-Schema Retention, and Active Hypothesis Discrimination in an Embodied World Model

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CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Kunal Lubhana: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project administration.

DECLARATION OF COMPETING INTEREST

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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DECLARATION OF GENERATIVE AI IN SCIENTIFIC WRITING

During the preparation of this manuscript, the author used generative AI models solely for editorial grammar refinement, stylistic readability polishing, and script formatting under rigorous, continuous human oversight. After using these assistance tools, the author independently reviewed and edited all content and takes full personal responsibility for the factual integrity and scientific claims of this publication.

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DATA AND CODE AVAILABILITY

The complete source code, 50-seed deterministic trajectory datasets, configuration files, and turnkey reproduction scripts (`reproduce_all.py`) associated with this study are openly available in the GitHub repository at <https://github.com/Kunallubhana77/DWMA-developmental-world-model-ai> and permanently preserved with open access in the Zenodo research archive at <https://doi.org/10.5281/zenodo.22796871>.